

Independent Study

Task 4 AIEA Lab | Apr. 23, 2026

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This Week

- Setting up and testing mini hypothesis
- Analysis and further action

Summary

This week I finished setting up my mini hypothesis and looked at some research on how things like cost efficiency are determined in the field. Next week I will finish setting up LTRAG's codebase and start setting things up for larger-scale experiments. I also need to do more literature review over the next few weeks.

Mini Experiment

This mini experiment taught me how to load FOLIO and ProofWriter datasets and carry out a meaningful small-scaled experiment. I tested an inference model (gpt-4.1-mini) against a reasoning model (gpt-5.4-nano) on real logic datasets using a baseline vs. a neuro-symbolic approach. Please refer to figure 1 under "References" to view the results. The code and results are also available [here](#) under nesy_reasoning/NeSyRe.

Analysis

We can tell that baseline inference models are better at FOLIO than reasoning models. However, the reasoning models seemed to perform better with ProofWriter data. ProofWriter allows for more reasoning-based NL language, so it makes sense why we're seeing something like that. The key analysis that proves my mini hypothesis, however, is that on FOLIO and ProofWriter, my inference model + neurosymbolic accuracy $\geq$ baseline reasoning accuracy.

Another interesting part here is that we have 21 errors. These errors are not many considering my pipeline has no fallback. LTRAG's error handling is what makes it superior, which is why it was my top choice. With LTRAG's error handling, neurosymbolic will beat reasoning models easier, and we can prove my larger hypothesis soon.

What now

Now I will keep setting up LTRAG and get it up and running by next week. I also will keep researching token tracking and finding valuable information to write about it. My next step for next week is to finish setting up LTRAG and FAISS RAG and run smoke tests and make sure it's working. Also, I should look up some valuable papers on cost/environmental analysis and efficiency.

References

AI Use:

I used Claude to help me better understand load_dataset and how to use it on Hugging Face datasets. I also used LINC's code as a reference.

Figure 1: Mini experiment stdout

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FOLIO benchmark (70 examples)

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FOLIO - gpt-4.1-mini | baseline: 70%

FOLIO - gpt-4.1-mini | neurosymbolic: 73%

FOLIO - gpt-5.4-nano | baseline: 51%

FOLIO - gpt-5.4-nano | neurosymbolic: 73%

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ProofWriter benchmark (70 examples)

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ProofWriter - gpt-4.1-mini | baseline: 46%

ProofWriter - gpt-4.1-mini | neurosymbolic: 97%

ProofWriter - gpt-5.4-nano | baseline: 61%

ProofWriter - gpt-5.4-nano | neurosymbolic: 86%

Errors: 21 ← Prover9 errors as a result of poor LLM NL to FOL translation

COT redirects: 0 ← this pipeline was disabled for this experiment.

Figure 2: graphed mini experiment

